

ECE 3337: Test #2

Spring 2022

Last name of student Solutions	Student ID number
First name of student	Email address

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	25	15	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	

1. (20 points) Laplace Transforms

- a. (6 points) A signal is given by $x(t) = t[u(t) - u(t - 2)] + (4 - t)[u(t - 2) - u(t - 4)] + 2\delta(t - 2)$. Find $\mathbf{X(s)}$, the Laplace transform of $x(t)$.

- (b) (4 points) Given a signal $x_2(t)$, find the Laplace transform of $y(t) = x_2(2t - 4)$ in terms of $\mathbf{X_2(s)}$

c. (10 points) Find the Laplace transform of $z(t) = 10 t e^{-2jt} \cos(7t) u(t)$, and identify its poles and zeros.

2. (20 points) Inverse Laplace Transform

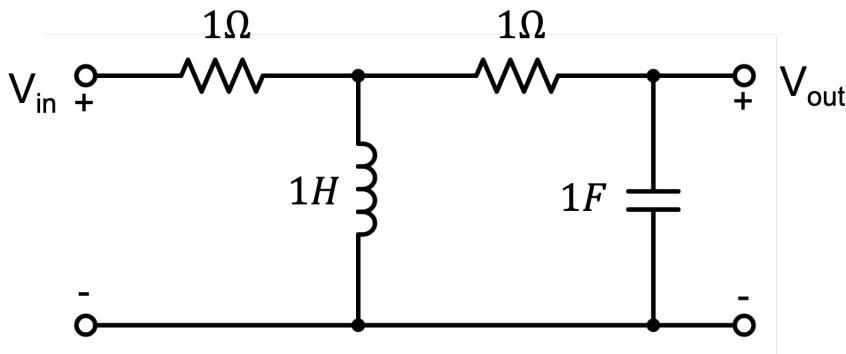
Find the inverse Laplace transforms for the following systems.

a. (10 points) $X(s) = \frac{e^{-s}s^2}{(s^2+2s+17)}$

b. (10 points) $X(s) = \frac{2s+1}{(s^3+2s^2+s)}$

3. (25 points) Circuit Analysis

For the circuit below, $V_{in}(t) = 2 \text{ V}$ for $t < 0$, and $V_{in}(t) = 2(1 - e^{-3t})u(t) \text{ V}$ for $t > 0$



a. (8 points) Draw the **s**-domain representation of the circuit.

b. (7 points) Find the **s**-domain representation of the input $V_{in}(t)$ for $t > 0$.

c. (10 points) Write mesh current equations in the s -domain that could be used to find the current through the capacitor. Use these equations to fill in the entries in the Cramer's method matrices shown below. Here, $\mathbf{I}_1$ and $\mathbf{I}_2$ represent the mesh currents, and $\mathbf{V}_1$, $\mathbf{V}_2$ represent voltage sources.

$$\begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{A}_3 & \mathbf{A}_4 \end{pmatrix} \begin{pmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{pmatrix}$$

Find the output $\mathbf{V}_{\text{out}}(s)$ in the Laplace domain.

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

$$\frac{d^2y(t)}{dt^2} + 10\frac{dy(t)}{dt} + 21 = \frac{dx(t)}{dt} + 2x(t)$$

The input is $x(t) = 10\sin(t)u(t)$, with the initial conditions $x(0^-) = 0, y(0^-) = y'(0^-) = 1$.

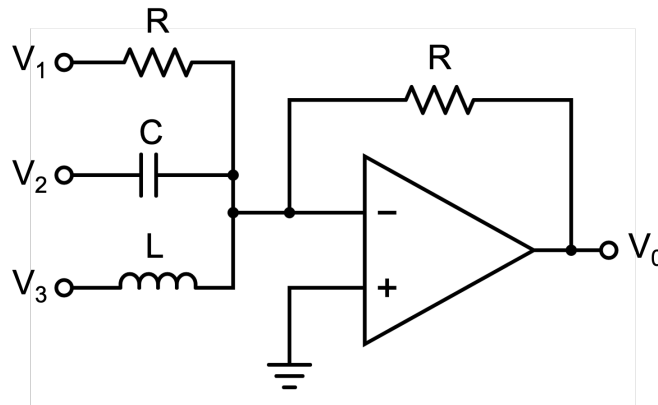
- a. (5 points) Derive a formula for the total output in the Laplace domain. Your answer should be expressed as a ratio of polynomials that are fully factored.

- b. (5 points) Calculate the natural response of this system in the Laplace domain

c. (5 points) Calculate the final value $y(\infty)$ of this system's response.

5. (10 points) Operational Amplifiers

For the circuit below, there was no energy stored initially in the capacitor or inductor.

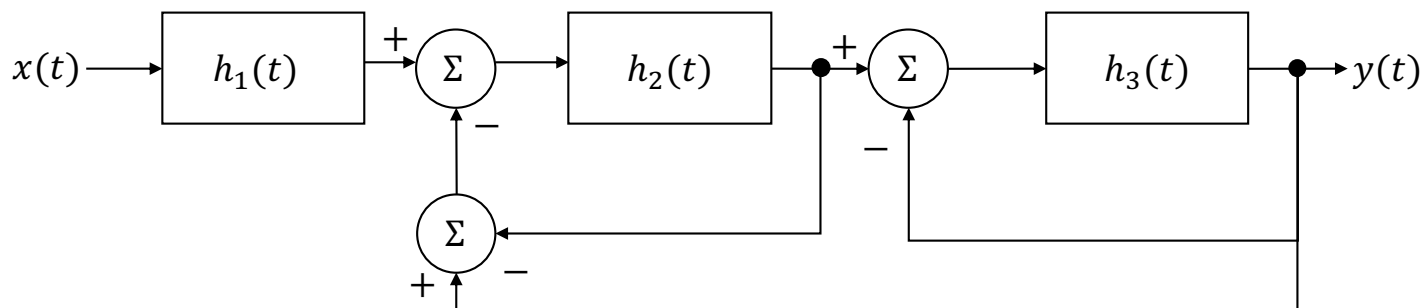


a. (5 points) Find the output voltage $V_0(s)$ in terms of input voltages $V_1(s)$, $V_2(s)$, and $V_3(s)$ and the circuit elements R , C and L .

b. (5 points) If $R = 1\Omega$, $L = 0.5H$, $C = 0.5F$, for what value of s will $V_0 = -(V_1 + V_2 + V_3)$?

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H(s)}$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



1(a)

$$\begin{aligned}x(t) &= t[u(t) - u(t-2)] + (4-t)[u(t-2) - u(t-4)] + 2\delta(t-2) \\&= t u(t) - t u(t-2) + (4-t) u(t-2) + (t-4) u(t-4) \\&\quad + 2\delta(t-2)\end{aligned}$$

$$\begin{aligned}&= t u(t) + (4-2t) u(t-2) + (t-4) u(t-4) \\&\quad + 2\delta(t-2)\end{aligned}$$

$$\begin{aligned}&= t u(t) - 2(t-2) u(t-2) + (t-4) u(t-4) \\&\quad + 2\delta(t-2)\end{aligned}$$

Taking Laplace transform on both sides

$$\begin{aligned}X(s) &= \frac{1}{s^2} - 2 \frac{1}{s^2} e^{-2s} + \frac{1}{s^2} e^{-4s} + 2e^{-2s} \\&= \boxed{\frac{1}{s^2} \left(\overset{\textcircled{A}}{1} - 2 \overset{\textcircled{B}}{e^{-2s}} + \overset{\textcircled{C}}{e^{-4s}} \right) + \overset{\textcircled{D}}{2e^{-2s}}}\end{aligned}$$

2 points for trying

1 each for $\textcircled{A}$ - $\textcircled{D}$

$$(b) \quad x_2(t) \xleftrightarrow{\mathcal{L}} X_2(s)$$

$$x_2(t-4) \xleftrightarrow{\mathcal{L}} X_2(s) e^{-4s} \quad \text{--- (A)}$$

1 point for trying
1 each for (A)-(C)

$$y(t) = x_2(2t-4) \xleftrightarrow{\mathcal{L}} \frac{1}{2} X_2\left(\frac{s}{2}\right) e^{-4\frac{s}{2}} = \frac{1}{2} X_2\left(\frac{s}{2}\right) e^{-2s}$$

(B) (C)

$$(c) \quad z(t) = 10t e^{-2jt} \quad \text{and} \quad u(t)$$

$$\cos(7t) u(t) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + 7^2} = \frac{s}{s^2 + 49} \quad \text{--- (A)}$$

$$e^{-2jt} \cos(7t) u(t) \xleftrightarrow{\mathcal{L}} \frac{s + 2j}{(s + 2j)^2 + 49} \quad \text{--- (B)}$$

$$10t e^{-2jt} \cos(7t) u(t) \xleftrightarrow{\mathcal{L}} 10 \frac{d}{ds} \left(\frac{s + 2j}{(s + 2j)^2 + 49} \right) \quad \text{--- (C)}$$

2 points for trying

1 each for (A)-(H)

$$= -10 \cdot \frac{((s + 2j)^2 + 49) - 2(s + 2j)(s + 2j)}{(s + 2j)^2 + 49}$$

$$= -10 \cdot \left(\frac{49 - (s + 2j)^2}{49 + (s + 2j)^2} \right) \quad \text{--- (D)}$$

$$= 10 \left[\frac{(s + 2j)^2 - 49}{((s + 2j)^2 + 49)^2} \right]$$

Poles : $(s+2j)^2 + 7^2 = 0$

$$(s+2j)^2 = -7^2$$

$$s+2j = \pm 7j$$

$$s = -2j \pm 7j = \boxed{-9j, +5j}$$

(E) (E)

Zeros : $(s+2j)^2 - 7^2 = 0$

$$s+2j = \pm 7$$

$$s = -2j \pm 7$$

$$= \boxed{-7-2j, +7-2j}$$

(A) (H)
repeated poles

$$-7, -2j, +7-2j$$

2. (20 points) Inverse Laplace Transform

Find the inverse Laplace transforms for the following systems.

a. (10 points) $X(s) = \frac{e^{-s}s^2}{(s^2+2s+17)} = e^{-s} \cdot X_2(s)$

Long division: $s^2+2s+17 \overline{) \begin{matrix} s^2 \\ s^2+2s+17 \\ -2s-17 \end{matrix}} \Rightarrow X_2(s) = 1 - \frac{(2s+17)}{(s+1)^2+4^2}$ (A)

$$s^2+2s+17 = (s+1)^2 + 4^2$$

$$= 1 - \frac{(2s+2+15)}{(s+1)^2+4^2}$$

$$= 1 - \frac{2(s+1)}{(s+1)^2+4^2} - \frac{15}{4} \cdot \frac{4}{(s+1)^2+4^2}$$
 (B)

$$\Rightarrow x_2(t) = \delta(t) - 2e^{-t} \cos(4t)u(t) - \frac{15}{4} \sin(4t)u(t)$$
 (C)

$$\Rightarrow x(t) = x_2(t-1) = \delta(t-1) - 2e^{-(t-1)} \cos(4(t-1))u(t-1) - \frac{15}{4} \sin(4(t-1))u(t-1)$$
 (D)

b. (10 points) $X(s) = \frac{2s+1}{(s^3+2s^2+s)}$

$$= \frac{2s+1}{s(s+1)^2} = \frac{A_1}{s} + \frac{B_1}{(s+1)} + \frac{B_2}{(s+1)^2}$$

$$A_1 = \left. \frac{2s+1}{(s+1)^2} \right|_{s=0} = 1$$
 (A)

$$B_2 = \left. \frac{(2s+1)}{s} \right|_{s=-1} = 1$$
 (B)

$$B_1 = \left. \frac{d}{ds} \left\{ \frac{2s+1}{s} \right\} \right|_{s=-1} = \left. \frac{d}{ds} \left\{ -\frac{1}{s^2} \right\} \right|_{s=-1} = -1$$
 (C)

$$\Rightarrow x(t) = u(t) - e^{-t}u(t) + te^{-t}u(t)$$
 (D)

Grading:

- 2 pts for trying
- +2 pts each for (A), (B), (C), (D)
- Full points for correct final answer

Grading:

- 2 pts trying
- +2 pts each for (A), (B), (C), (D)
- Full points for correct final answer

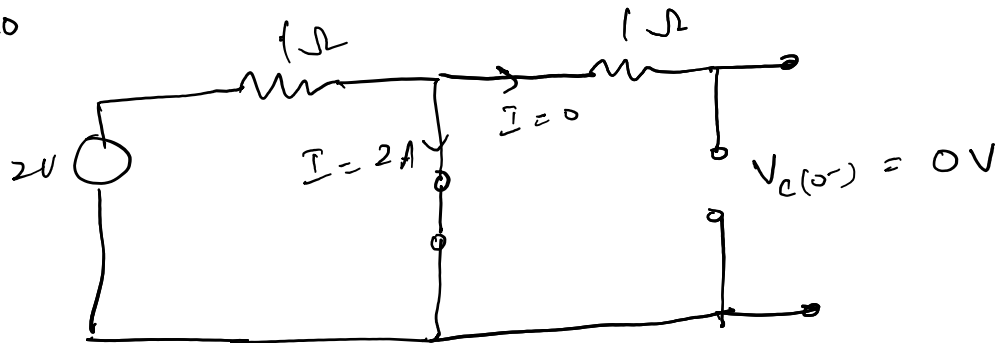
3(a)

2 points for trying
1 each for (A)-(F)

$$V_{in} = 2 \text{ V for } t < 0$$

Find initial conditions

for $t < 0$



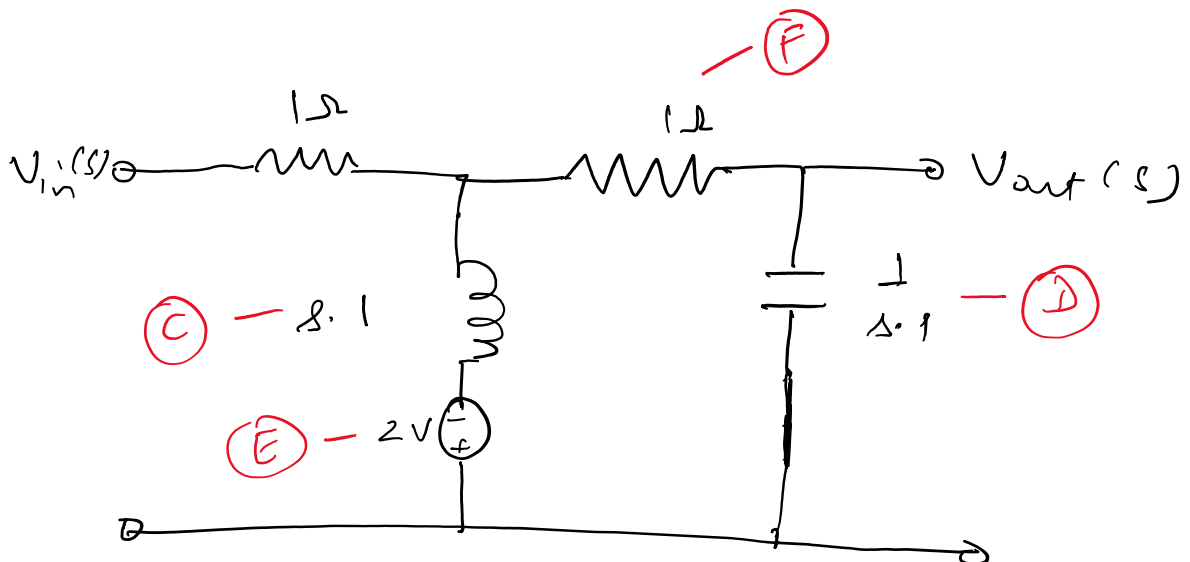
$$i_L(0^-) = \frac{2 \text{ V}}{1 \Omega} = 2 \text{ A}$$

(A)

$$V_C(0^-) = 0.2 \text{ A} \times 1 \Omega = 0 \text{ V}$$

(B)

S-domain representation of the circuit is



(b) $v_{in}(t) = 2(1 - e^{-3t}) u(t)$ for $t > 0$

2 points for
trying
1 each for

(A) - (E)

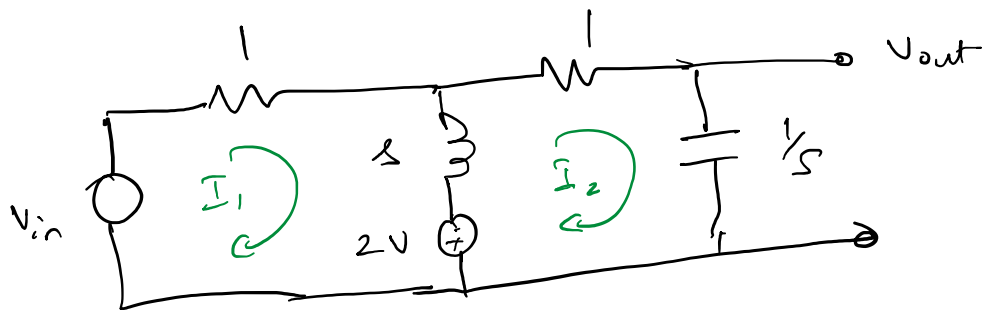
$$V_{in}(s) = \mathcal{L}[v_{in}(t)] \quad \text{(A)} \quad \text{(B)}$$

$$= \mathcal{L}[2u(t)] - \mathcal{L}[2e^{-3t}u(t)]$$

$$= \frac{2}{s} \quad \text{(C)} \quad \text{(D)} - \frac{2}{s+3}$$

$$= 2 \frac{3}{s(s+3)} = \frac{6}{s(s+3)} \quad \text{(E)}$$

(c)



using
KVL

2 points
for trying
1 each for
(A) - (F)

2 points for
(G)

$$\text{(A)} \quad v_{in} - I_1 - s(I_1 - I_2) + 2V = 0$$

$$\text{(B)} \quad s(I_2 - I_1) + 2 + \frac{I_2}{s} + I_2 = 0$$

$$(1+s)I_1 - sI_2 = v_{in} + 2$$

$$-sI_1 + \left(\frac{1}{s} + 1 + s\right)I_2 = -2$$

$$\textcircled{C} \begin{bmatrix} (1+s) & -s \\ -s & (1+s+\frac{1}{s}) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_{in} + 2 \\ -2 \end{bmatrix}$$

Using Cramer's method

$$\Delta = \begin{vmatrix} 1+s & -s \\ -s & 1+s+\frac{1}{s} \end{vmatrix} = (1+s)(1+s+\frac{1}{s}) - s^2 \quad \textcircled{E}$$

$$\Delta_{I_2} = \begin{vmatrix} 1+s & V_{in} + 2 \\ -s & -2 \end{vmatrix} = -2(1+s) + s(V_{in} + 2) \quad \textcircled{F}$$

$$I_2 = \frac{\Delta_{I_2}}{\Delta}$$

$$V_{out} = \frac{1}{s} \cdot I_2 = \frac{\Delta_{I_2}}{\Delta}$$

$$= \frac{-2(1+s) + s(V_{in} + 2)}{s \left[(1+s) \left(1+s+\frac{1}{s} \right) - s^2 \right]}$$

Substituting $V_{in} = \frac{6}{s(s+3)}$

$$= \frac{-2(1+s) + \left(\frac{6}{s(s+3)} + 2\right)s}{(1+s)(s+s^2+1) - s^3}$$

$$= \frac{-2(1+s) + 2\left(3 + s^2 + 3s\right) \cancel{s} \cancel{(s+3)}}{(1+s)(s+s^2+1) - s^3}$$

$$= \frac{- (1+s)(s+3) + (3 + s^2 + 3s)}{(s+3) \left[(1+s)(s+s^2+1) - s^3 \right]}$$

6

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

$$\frac{d^2 y(t)}{dt^2} + 10 \frac{dy(t)}{dt} + 21y(t) = \frac{dx(t)}{dt} + 2x(t)$$

The input is $x(t) = 10\sin(t)u(t)$, with the initial conditions $x(0^-) = 0, y(0^-) = y'(0^-) = 1$.

- a. (5 points) Derive a formula for the total output in the Laplace domain. Your answer should be expressed as a ratio of polynomials ~~that are~~ ^{with} fully factored denominator.

$$[s^2 Y(s) - sy(0^-) - y'(0^-)] + 10[sY(s) - y(0^-)] + 21Y(s) = sX(s) - x(0^-) + 2X(s)$$

$$[s^2 Y(s) - s - 1] + 10[sY(s) - 1] + 21Y(s) = sX(s) + 2X(s) \quad (A)$$

$$Y(s)[s^2 + 10s + 21] = (s+2)X(s) + s + 11$$

$$\text{Now, } X(s) = \frac{10}{s^2+1} \quad (B) \text{ \& } s^2 + 10s + 21 = (s+3)(s+7) \quad (C)$$

$$\Rightarrow Y(s) = \frac{10(s+2)}{(s^2+1)(s+3)(s+7)} + \frac{s+11}{(s+3)(s+7)} \quad (D)$$

$$= \frac{10(s+2) + (s^2+1)(s+11)}{(s^2+1)(s+3)(s+7)} = \frac{s^3 + s^2 + 11s + 21}{(s^2+1)(s+3)(s+7)} //$$

Grading: • 1 pt for trying
• +1 pt each (A) (B) (C) (D)

- b. (5 points) Calculate the natural response of this system in the Laplace domain

The natural response is associated with the poles $\{3, 7\}$

For this, it is sufficient that we calculate only the 2 relevant terms $\frac{A_1}{(s+3)}$, $\frac{A_2}{(s+7)}$ in the partial fraction expansion for $Y(s)$.

Grading:

• 1 pt trying
• +1 pt each (A) (B)
• +2 pt (C)

$$A_1 = -\frac{3}{4} \quad (A) \quad A_2 = \frac{7}{4} \quad (B) \Rightarrow Y_{\text{natural}}(s) = -\frac{3}{4(s+3)} + \frac{7}{4(s+7)} //$$

- c. (5 points) Calculate the final value $y(\infty)$ of this system's response.

The partial fraction expansion of $Y(s)$ is of the form

$$\frac{A_1}{(s+3)} + \frac{A_2}{(s+7)} + \underbrace{\frac{B_1}{(s^2+1)} + \frac{B_2 s}{(s^2+1)}}$$

we see that the output contains undamped oscillations

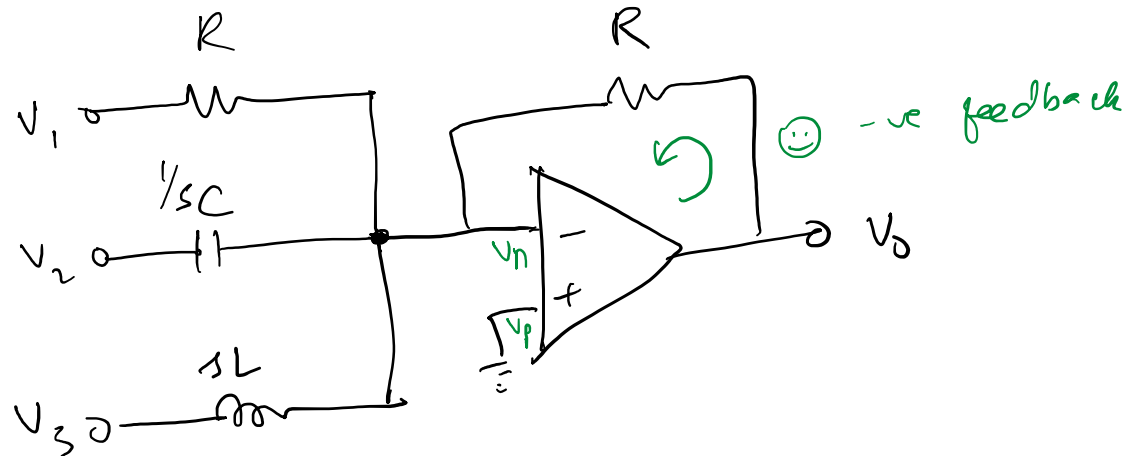
$\Rightarrow$ the final value $y(\infty)$ is indeterminate

Grading:

- 2 pts trying, but incorrect answer
- Full pts for correct answer

1 point for trying, 1 each for (A) (D)

5(a)



- There is $-ve$ feedback — (A)
- $V_p = 0$, $V_p = V_n = 0$ — (B)
- Current through R , C and L are

$\frac{V_1}{R}$, $\frac{V_2}{1/sC}$, $\frac{V_3}{sL}$ respectively

$$V_0 = V_n - I R$$

$$I = \frac{V_1}{R} + sC V_2 + \frac{V_3}{sL} \quad \text{— (C)}$$

$$V_0 = - \left(\frac{V_1}{R} + sC V_2 + \frac{V_3}{sL} \right) R \quad \text{(D)}$$

$$= - \left(V_1 + sRC V_2 + \frac{R}{sL} V_3 \right)$$

(b) If $R = 1 \Omega$, $L = 0.5 H$, $C = 0.5 F$,

$$V_0 = - \left(V_1 + 0.5s V_2 + \frac{1}{0.5s} V_3 \right) \quad \textcircled{A}$$

For $\boxed{S = 2}$, $V_0 = -(V_1 + V_2 + V_3)$

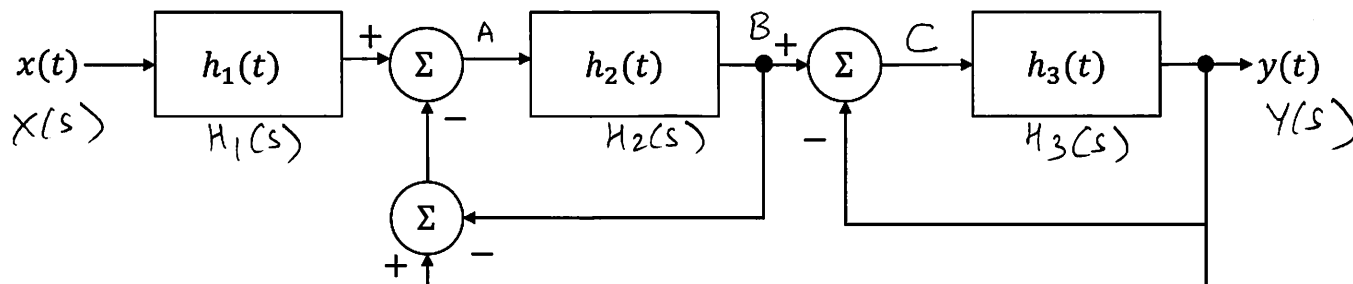
$\textcircled{B}$

1 point for trying

2 each for $\textcircled{A}$ and $\textcircled{B}$

6. (10 points) System Block Diagrams

Calculate the overall transfer function $H(s)$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



Using simplified notation (drop "(s)")

$$A = XH_1 + B - Y \quad \textcircled{A}$$

$$C = B - Y \quad \textcircled{B}$$

$$Y = H_3 C \quad \textcircled{C}$$

$$B = H_2 A \quad \textcircled{D}$$

$$\Rightarrow A = XH_1 + H_2 A - Y \quad \Rightarrow A(1 - H_2) = XH_1 - Y$$

$$Y = H_3 (B - Y) = H_3 (H_2 A - Y)$$

$$\Rightarrow Y(1 + H_3) = H_3 H_2 A$$

$$= H_3 H_2 \frac{(XH_1 - Y)}{1 - H_2}$$

$$\Rightarrow Y \left(1 + H_3 + \frac{H_3 H_2}{1 - H_2} \right) = \frac{X H_1 H_2 H_3}{1 - H_2}$$

$$\Rightarrow \frac{Y}{X} = H = \frac{H_1 H_2 H_3}{(1 + H_3)(1 - H_2) + H_3 H_2} = \frac{H_1 H_2 H_3}{(1 + H_3 - H_2)} \quad \textcircled{E}$$

Grading:

- 2 pts trying
- + 1 pt each
- $\textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}$
- + 2 pt $\textcircled{E}$